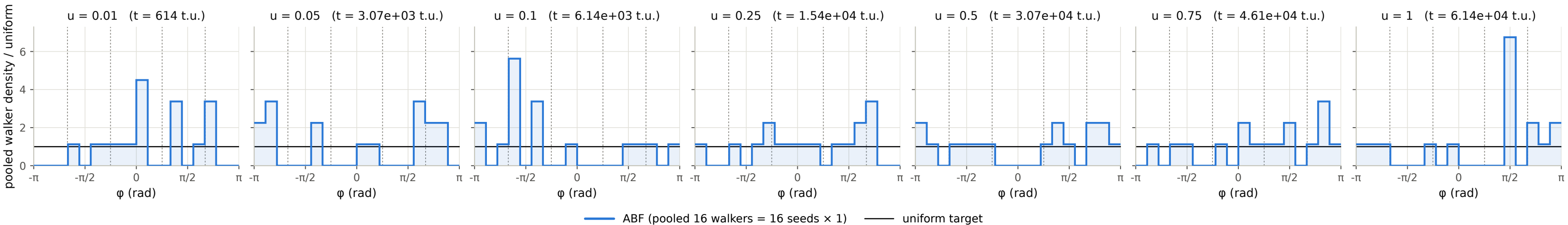


Ethane/LTA 150 K · N = 1 · T_N = 61440 t.u.

n_steps = 307,200,000 · h = 0.0002 · B = N n_steps = 3.072e+08 walker-steps · complete seeds: ABF 16/16 (N = 1: ABF only, no FR arm)

Instantaneous marginal snapshots (seed-pooled), ABF vs ABF+FR



Seed-pooled instantaneous histograms of the walkers at the snapshot u (the saved state), on the 18 coarse bins of the TV metric; density relative to the uniform target (1 = uniform). Not a median: pooled counts.

Small N (N = 1, i.e. ≤ 4): each seed contributes only 1 walker to an instantaneous histogram, so instantaneous marginals are necessarily discrete (TV_inst floor $E_N[TV]$ = 0.944).

Dashed guides: $|\phi| = 0.791$ rad (window $|z| < 1.5$ Å) and $|\phi| = 2.109$ rad (cage $|z| > 4$ Å), $z = |\phi| a / 2\pi$, $a = 11.919$ Å.